Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Dr. Okan Avci	Email	okan.avci@ipa.fraunhofer.de
Institute Name	Fraunhofer IPA			
Address	Nobelstr. 12, 70569 Stuttgart, Germany			

2. Technology Overview

Title of Proposed Technology	In silico technologies for virtual trials of shoulder endoprostheses using realistic virtual human models			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Advanced Bio		Advanced Bio	
	Medium Category	Small Category	Medium Category	Small Category
	Bio-processes / Bio-Equipment	Bio-engineering Technologies	Functional Restoration / Assistive & Welfare Devices	Implants
Concept of Proposed Technology	Joint arthroplasty is currently guided by geometric analysis of static joint positions using medical imaging. However, the effects of alignment, as well as the functional constraints and limitations of endoprostheses on the musculoskeletal system, are not fully understood [1]. This is true for all joints, but particularly for the shoulder joint. The shoulder joint is primarily stabilized by the strong muscle-tendon unit (MTU) and ligaments [2]. Shoulder injuries are increasing due to heavy physical work, aging population and sports accidents. Surgical treatment of shoulder injuries involving the MTU is sophisticated and can even potentially worsen shoulder mechanics. The biomechanics of the shoulder joint system is complex. Disturbances and overstress caused by implants and/or surgery can result in muscle imbalances and unnatural movements, leading to pain [3]. Currently, there is no analysis tool for the virtual planning of shoulder interventions on the MTU that evaluates and optimizes the positioning of endoprostheses for dynamic arm motion, considering			

	muscle force generation, physiology, and the anatomy of the human shoulder joint. Existing multi-body simulations of the shoulder do not incorporate implants, and one-dimensional model reduction is inadequate for simulating contractile stress loading and the deformation behavior of tissues [4]. Additionally, the 1D simulation approach oversimplifies the musculoskeletal system to such an extent that the results are not reliable. We have been developing a cutting-edge in silico human modeling and simulation platform for orthopedic applications targeting the ankle, knee, hip, elbow, and shoulder joints [6, 7]. This platform allows us to investigate and validate endoprostheses in a realistic 3D physiological-anatomical manner, utilizing muscle-force-driven joint kinematics. The physiology and anatomy of both soft and hard tissues are crucial for our continuum-mechanical modeling approach in numerical simulations using the finite element method (FEM). These models are derived from medical imaging and muscle physiology experiments, which are already integrated into our finite element (FE) computational models [5, 6]. Our in silico technologies also enable the development of next-generation patient-specific biomechanical endoprostheses. These new endoprostheses could function more biomechanically effectively while harmonizing better with the human musculoskeletal system. Additionally, innovations in endoprostheses can be tested through in silico trials on a cohort of virtual patients, significantly reducing the costs associated with clinical trials required for the approval process. Clinical trials account for over 50% of the costs of implant development [8], and in silico trials could reduce these costs by more than 30% and shorten the time to market by approximately 2-3 years [9].
Rationale for Joint R&D Project	In silico trials on orthopedic implants using physiological and anatomical human models are essential for providing more evidence in human-centered product design and virtual surgery planning, ultimately leading to better decision-making. With realistic virtual patients, critical decisions can be supported to enhance patient-specific treatment. In real patients, such studies are not feasible, as surgeons typically have only one opportunity to implant an endoprosthesis for each patient, leaving no room for variation in study. Our goal is to illuminate this "black box" for greater evidence in orthopedic medicine through in silico technologies. To account for population variability in age, anthropometry, and physiological conditions, we seek collaboration with Korean SMEs and hospitals to design in silico technologies tailored to the needs of the Korean population. Additionally, the prevalence of orthopedic diseases can vary from nation to nation, making it crucial to consider country-specific treatments for shoulder diseases and the product portfolio of implant manufacturers in Korea. This requires a close collaboration with Korean institutions and SMEs.
Key Words	• MRI-based semi-automated segmentation of soft tissue structures and muscle fibers to create 3D musculoskeletal FE computational models, including the connective tissues of the shoulder. • Forward-dynamics simulations of physiological-anatomical musculoskeletal joint systems, incorporating joint diseases and treatments using FEM. • In silico studies of endoprostheses utilizing virtual patient models to evaluate biomechanical function and optimize designs for individual patients.

	• In silico investigations of surgical techniques to demonstrate their impact on implant performance and patients' biomechanical system. • In silico trials with a virtual patient cohort to support implant clinical studies.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	We will develop cutting-edge technologies for in silico trials, enabling us to create physiologically and anatomically realistic virtual cohorts of patients specific to the Korean population. Using these models, we will establish a simulation process for the human-centered virtual development of endoprostheses and enable their virtual clinical approval. In this R&D project, we aim to explore various industrial applications for virtual human models. Our key objectives are as follows: • **Proof of Physiological Function:** Validate the physiological function of endoprostheses through forward-dynamic musculoskeletal simulations, which cannot be feasibly conducted experimentally in real-life settings. • **Surgeon Impact on Decision-Making:** Assess how a surgeon's decisions regarding implant positioning and surgical interventions on soft tissue can influence muscle force generation and joint kinematic performance. Specifically, the unique positioning of reverse shoulder endoprostheses can significantly reduce the performance of the deltoid muscle while increasing moment forces [3]. • **In Silico Trials of Endoprostheses:** Conduct in silico trials of various endoprostheses on a virtual cohort of patients to evaluate safety (including structural stability and realistic joint contact force for wear study) and biomechanical efficiency. These assessments are currently unfeasible in clinical trials, as each endoprosthesis can only be implanted once per patient.			
R&D Target Specifications	The objectives are organized into three categories: 1. **Computational Models:** Development of physiological shoulder models for a cohort of virtual patients within a verification and validation (V&V) framework. 2. **Performance Study of Endoprostheses:** Evaluation of the biomechanical function of endoprostheses and the impact of surgical decision-making on muscle imbalances resulting from muscle or tendon transfer or removal. 3. **In Silico Study:** Quantification of the harmful effects of overstressing the human body through interventions, aiming to identify the best possible solutions. We will demonstrate the predictive quality of the shoulder models defined in the context of use (CoU) for this project. We will adhere to the standard guidelines for the credibility of computational modeling and simulation (CM&S) established by the FDA [10], which identifies in silico clinical trials as future technologies for enhancing patient-specific product performance, ensuring patient safety, and reducing R&D costs for implants.			

	Additionally, we will analyze how the performance of different designs of endoprostheses within a muscle-force-driven musculoskeletal shoulder model varies from patient to patient. This will provide implant manufacturers with objective support for designing their products while considering patients' anatomical, physiological, and biomechanical conditions. We also aim to identify the impact of the surgeon's decision-making on the performance of endoprostheses, such as the positioning of the implant and interventions on muscles, tendons, and connective tissues, as these factors significantly affect muscle force generation.

R&D Plan	Our detailed plan to achieve the objectives of this R&D project is summarized as follows:
	Part 1
	• Record MRI scans of the shoulder and torso of 10 subjects selected to reflect a relevant population distribution in a hospital in Korea. We will assist the hospital in planning the recording sessions. Biomechanical studies, including motion capture and muscle physiology assessments (EMG), will be conducted for these subjects. The personal data collected will serve as input for CM&S. For this purpose, we will require an ethical proposal, which must be submitted by a clinician from a partner hospital in Korea.
	• The muscles will be individually contracted to generate the force necessary for arm motion around the shoulder joint. Muscle activation patterns over time will be determined using a neuronal feedforward network.
	• Based on the collected data, we will create a digital twin of the musculoskeletal system for each subject, which will serve as the primary database for the virtual patient cohort.
	Milestone 1: In silico shoulder models of a cohort of selected subjects
	
	Part 2
	• In this project, we will develop three use cases for an abduction study of the shoulder up to 90°, which are crucial for evaluating shoulder arthroplasty. The use cases are:
	  ○ Healthy shoulder muscles (reference case) of the subjects,
	  ○ Torn rotator cuff and
	  ○ Variation in the size and positioning of two endoprostheses within the shoulder.
	• Both types of endoprostheses from the Korean SME that we will investigate in this project should already be in use, allowing for comparisons between them and with the current available data and expertise regarding endoprostheses. If the SME has a new prototype of a shoulder implant, it can also be analyzed and compared within this project.
	• We will compare the stress loading on the subjects' musculoskeletal systems in both healthy and diseased states, as well as with the endoprostheses, by simulating and studying the defined use cases.
	Milestone 2: Virtual study of the biomechanical aspects of endoprostheses
	
	Part 3:
	• **V&V Process:** The V&V process encompasses the anatomy and physiology of the musculoskeletal shoulder system of 10 subjects, ensuring the credibility of the human models and simulations for the approval process of in silico clinical trials [10].
	• **Evaluation of Endoprosthesis Wear:** Study and evaluation of endoprosthesis wear during

	abduction, considering contact pressure. The wear of metal or polymer materials is critical for patients, as material particles can cause inflammation and pain, often necessitating a revision of the endoprosthesis. Each surgical intervention increases the risk of complications, and the probability of failure for the second implant also rises due to the need for new preparation of the musculoskeletal joint. • **Assessment of Endoprostheses:** Assessment and classification of two different endoprostheses based on size, position, and patient-specific shoulder biomechanics using the virtual cohort of patients. • **Guideline Development:** Development of guidelines for virtual surgical planning of endoprostheses, taking into account variations within the virtual cohort related to muscle physiology, muscle force generation, shoulder anatomy, and other relevant features. **Milestone 3: Optimizing the performance of various endoprostheses and developing surgical guidelines to enhance patient-specific decision-making.**

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Our role in this project is to lead the research and development of a new in silico platform. This platform will use realistic virtual human models to evaluate endoprostheses and conduct virtual clinical trials. The aim is to enhance the evidence base and reduce the time required for clinical studies. We will manage the project to ensure that the timeline for deliverables and milestones is met. We will maintain communication with our Korean partners, including implant manufacturers and the hospital, from which we will obtain medical imaging and EMG data of the subjects. For data security reasons, we will work with anonymized data. We will collaborate closely with the Korean SMEs and the hospital. We will hold online meetings and visit them at regular intervals. We will also invite the Korean SMEs to visit us. The subjects are solely needed for data acquisition to develop the virtual cohort of patients. We will not consider medical treatments in this project and will not use the patients for post-operative validation procedures. The subjects may also include individuals without any shoulder injuries or pain. We can implement various musculoskeletal shoulder joint diseases into virtual models based on any subject. There is no need for post-operative V&V of patients, as this project does not involve clinical validation of long-term patient outcomes. Our expertise in this project focuses on CM&S to study the biomechanics of the locomotor system and develop a specific in silico technology platform. You will also gain insights into our previous and ongoing R&D in in silico technologies for human modeling of various joints, implants, and virtual planning for transtibial amputee surgery and prosthetic limb sockets. We will utilize our Linux workstation cluster for high-performance computing when conducting in silico trials on the virtual cohort of patients for the endoprostheses.

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Expected R&R of Korean SME Partner	In this project, close cooperation with the Korean partners is essential to ensure successful outcomes, which depend on the quality of the data acquired from both the SMEs and the hospital. We will request construction information for 2-3 different endoprostheses from the Korean SMEs, including CAD data, material specifications, functional information, and surgical guidelines for implant positioning within the shoulder. If the SME is currently developing a new implant, we will support and optimize the prototype using the virtual cohort of patients. The SME implant manufacturer will provide current performance assessments of their implants, serving as a solid starting point for comparison and expansion of the dynamic evaluation through in silico analysis. Therefore, we will maintain regular communication via online meetings and, as mentioned, hold in-person meetings in Korea and Germany at appropriate intervals. We need the SME implant manufacturer to provide a detailed description of the needs, requirements, and challenges related to the R&D of shoulder implants, as well as their wishes and expectations for the in silico trials as future technology. We expect SMEs to maintain an open and positive outlook, while also offering constructive feedback on our new approach to the virtual testing of endoprostheses using virtual patients. In a successful project outcome, we hope the SMEs will embrace the challenges of implementing the new in silico technologies within the SME partners, supported by our guidance and training. For this R&D project, we will require a clinical partner to recruit Korean adult subjects, specifically an equal number of 5 women and 5 men aged 20-50. The clinic must be equipped to conduct biomechanical motion and EMG experiments on the subjects, as well as obtain MRI images. The protocol for the MRI will be provided by us.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	In this project, we will develop guidelines for manufacturers and surgeons on how to optimize the biomechanical performance of shoulder endoprostheses. We will assess the stress on the musculoskeletal system due to the intervention and evaluate the potential risk of damage. We have extensive expertise in in silico technologies and methodologies for creating a virtual patient cohort for shoulder treatment studies. In this project, we will develop new insights for our Korean partners on how endoprostheses work and perform biomechanically within the patient's body. Our goal is to uncover previously unknown correlations that can lead to better patient treatment and improved product development. The manufacturer could license the novel in silico technologies for further internal use within their company. We will train Korean SMEs to use our computational software for the continuous improvement of their implants using virtual human models. This will enable the Korean companies to engage in in silico trials, reducing R&D costs, shortening the duration of clinical trials, and accelerating the time-to-market process. Through our collaboration, we can commercialize in silico technologies for the analysis of implants using the developed virtual cohort of Korean subjects. We can then offer these technologies to other implant manufacturers in Korea and East Asia for their own R&D, as well as for analyzing and

	optimizing shoulder endoprostheses. The impact will be substantial, as the current development procedures for endoprostheses rely on labor-intensive mechanical testing or cadaveric tests, which do not accurately represent real human conditions. In silico technologies utilizing virtual human models represent the future of evidence-based medicine by enhancing product development through digital twins of humans and accelerating clinical trials. Creating a wider variety of basic virtual cohorts will increase the patient population and help identify physics-based flaws in prototypes, thereby reducing the costly iteration process in product development. Additionally, the use of in silico trials to minimize animal testing. There are many compelling reasons to believe that in silico technologies will succeed, as there is no viable alternative to virtual human models in the future.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization